

Anatomy of a Cohomology Wall

The Koszul section-map rank defect on the tetraquadric Calabi–Yau threefold

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Abstract

Line-bundle cohomology on Calabi–Yau threefolds is piecewise polynomial in the line-bundle charges, with the polynomial pieces separated by walls [2]; the values are computable by established tools [4], and proofs of such formulae exist for surfaces and certain threefolds via vanishing-theorem and Zariski-chamber techniques [5, 6]. This note dissects the *mechanism* of one wall family on the tetraquadric $X = [(\mathbb{P}^1)^4 \parallel (2, 2, 2, 2)]$, for charges $D = (-a_1, -a_2, -a_3, b)$ with $a_i \geq 4$ and $b \geq 2$.

We prove a closed-form rank formula for the Koszul connecting map at $b = 2$, and we prove that the rank defect saturates at the constant $48 = 3! \cdot 2^3$ for all $a_i \geq 6$, by two independent arguments: a finite-difference identity on the Koszul complex, and an identification of the cokernel with the cohomology of a base-locus scheme of degree $\deg Z = \int_Y c_1(\mathcal{O}(2, 2, 2, 2))^3 = 48$. The wall structure reflects two distinct piecewise phenomena: boundary H^1 corrections that activate where individual ambient H^1 groups switch on, and a separate source-target minimum flip. The rank formula is verified against 22 independently computed charge vectors (22/22 exact).

For $b \geq 3$ the section map reduces to a genuinely partial contraction. We prove a general Toeplitz determinantal theorem for every $b \geq 2$: the maximal-minor ideal of the section-induced Toeplitz map is exactly the $(b - 1)$ st power of the length-48 base-locus ideal, the local cokernel has length $48 \binom{b+1}{3}$, and in the acyclic range $a_i \geq 2b + 1$ the global rank defect is given by an explicit source-target formula. The Thom–Porteous ceiling occurs precisely when $(b + 1) \prod_i (a_i - 1) \geq (b - 1) \prod_i (a_i + 1)$. Exact finite-field computations at $b = 4$ and $b = 5$ become verification data for this theorem and rule out the competing factorial pattern.

1 Setup

The tetraquadric $X = [(\mathbb{P}^1)^4 \parallel (2, 2, 2, 2)]$ is the smooth zero locus of a generic section $s \in H^0(A, \mathcal{O}_A(N))$ of $N = (2, 2, 2, 2)$ on the ambient space $A = (\mathbb{P}^1)^4$. It is a Calabi–Yau threefold with $(h^{1,1}, h^{2,1}) = (4, 68)$, $\chi = -128$. For a line bundle $L = \mathcal{O}_X(D)$, $D \in \mathbb{Z}^4$, cohomology is computed from the Koszul sequence

$$0 \longrightarrow \mathcal{O}_A(D - N) \xrightarrow{-s} \mathcal{O}_A(D) \longrightarrow \mathcal{O}_X(D) \longrightarrow 0. \quad (1)$$

Ambient groups factor through Künneth from the $\mathbb{P}^1$ atoms,

$$h^0(\mathbb{P}^1, \mathcal{O}(k)) = \max(k + 1, 0), \quad h^1(\mathbb{P}^1, \mathcal{O}(k)) = \max(-k - 1, 0). \quad (2)$$

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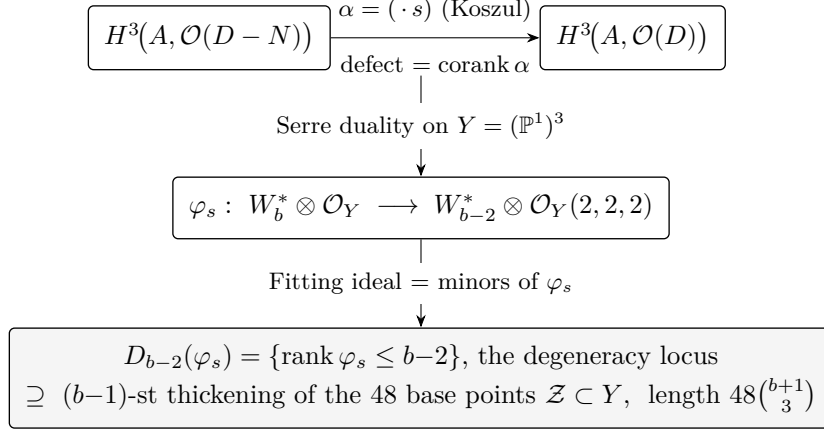


Figure 1: The mechanism in one picture. The connecting Koszul section map α on the ambient $A = (\mathbb{P}^1)^4$ dualises, by Serre duality on $Y = (\mathbb{P}^1)^3$, to a contraction φ_s of vector bundles of ranks $(b+1) \rightarrow (b-1)$. The section-induced Toeplitz form is now computed exactly for every $b \geq 2$: its maximal-minor ideal is the $(b-1)$ st power of the 48-point base-locus ideal, and its local cokernel length is $48\binom{b+1}{3}$ (Theorem 5.1). After twisting in the acyclic range, the global defect is governed by the same length together with the source-target minimum.

The long exact sequence of (1) contains the connecting multiplication map

$$\alpha : H^q(A, \mathcal{O}_A(D-N)) \longrightarrow H^q(A, \mathcal{O}_A(D)), \quad (3)$$

in the degree q where both groups are nonzero (the *section regime*).

Definition 1.1 (Rank defect). $\text{def}(D) := \min(\dim_{\text{src}}, \dim_{\text{tgt}}) - \text{rank } \alpha \geq 0$, where src and tgt are the source and target of (3). A bundle is *defective* when $\text{def}(D) > 0$.

2 The family and the contraction mechanism

Fix the sign pattern with three negative charges,

$$D = (-a_1, -a_2, -a_3, b), \quad a_i \geq 4, \quad b \geq 2,$$

so that $q = 3$ in (3): the three negative factors carry H^1 , the fourth carries H^0 . Write $a = (a_1, a_2, a_3)$ and let $Y = (\mathbb{P}^1)^3$ carry the unnamed cohomology below.

Proposition 2.1 (Reduction; explicit dimensions). *[proven] Set $W_k = H^0(\mathbb{P}^1, \mathcal{O}(k))$. By Künneth and (2),*

$$\dim_{\text{src}} = h^3(A, \mathcal{O}(D-N)) = (b-1) \prod_{i=1}^3 (a_i+1), \quad \dim_{\text{tgt}} = h^3(A, \mathcal{O}(D)) = (b+1) \prod_{i=1}^3 (a_i-1).$$

Serre duality on A (with $K_A = \mathcal{O}(-2, -2, -2, -2)$), under which multiplication by s is self-adjoint, identifies $\text{rank } \alpha$ with the rank of the partial contraction

$$\tilde{\mu} : H^0(Y, \mathcal{O}(a-2)) \otimes W_b^* \longrightarrow H^0(Y, \mathcal{O}(a)) \otimes W_{b-2}^*, \quad (4)$$

where $s = \sum_m s_m \otimes y^m$ (y^m a monomial basis of W_b , slices $s_m \in H^0(Y, \mathcal{O}(2, 2, 2))$) acts by $\xi \otimes \lambda \mapsto \sum_m (s_m \xi) \otimes (\lambda \lrcorner y^m)$. For $b = 2$ the residual factor W_0^ is trivial and (4) collapses to the multiplication map*

$$\mu : U \otimes H^0(\mathcal{O}(a-2)) \longrightarrow H^0(\mathcal{O}(a)), \quad U = \langle s_1, s_2, s_3 \rangle \subset H^0(\mathcal{O}(2, 2, 2)). \quad (5)$$

Proof. The Künneth dimensions follow from (2): for $\mathcal{O}(D)$ the unique degree-3 contribution takes H^1 on the three negative factors $(\dim a_i - 1)$ and $H^0(\mathcal{O}(b))$ $(\dim b + 1)$; for $\mathcal{O}(D - N)$ the same with $-a_i - 2$ $(\dim a_i + 1)$ and $\mathcal{O}(b - 2)$ $(\dim b - 1)$. Dualizing α and applying Serre duality factorwise, $H^1(\mathbb{P}^1, \mathcal{O}(k))^* \cong H^0(\mathbb{P}^1, \mathcal{O}(-k - 2))$, turns each negative factor's H^1 into polynomial spaces and the fourth factor's $H^0(\mathcal{O}(b))$, $H^0(\mathcal{O}(b - 2))$ into their duals; multiplication by s becomes the stated contraction. Self-adjointness of multiplication under the Serre pairing gives $\text{rank } \alpha = \text{rank } \tilde{\mu}$. $\square$

Proposition 2.2 (Dichotomy). *For generic s :*

1. **[proven]** $b = 2$. U is spanned by a regular sequence (Lemma 3.1); its base scheme $\mathcal{Z} \subset Y$ is finite of length $\deg \mathcal{Z} = \int_Y (2H_1 + 2H_2 + 2H_3)^3 = 3! \cdot 2^3 = 48$.
2. **[proven]** **General $b \geq 2$ Toeplitz saturation in the acyclic range.** The reduction is the genuinely partial contraction (4). The section-induced Toeplitz maximal-minor ideal and the Buchsbaum–Rim cokernel length are computed for all $b \geq 2$, and for $a_i \geq 2b + 1$ the global defect is the piecewise source-target formula of Theorem 5.1.
3. **[proven]** **2+2 pattern $(-a_1, -a_2, b_1, b_2)$.** For $b_1 = b_2 = 2$, the contraction is multiplication by the 9-dimensional linear system $V \subset H^0((\mathbb{P}^1)^2, \mathcal{O}(2, 2))$ cut out by the slices of s . Since $\mathcal{O}(2, 2)$ is very ample on $(\mathbb{P}^1)^2$ and three generic sections of a very ample bundle on a surface already have empty common zero locus (expected codimension $3 > 2$), the base scheme of V is empty [8, III.10.9]. Emptiness alone gives only surjectivity of the sheaf map; surjectivity on H^0 (hence $\text{def} = 0$) follows because the kernel sheaf, resolved by the Koszul complex of the slices, has vanishing H^1 : on the surface $(\mathbb{P}^1)^2$ every relevant twist $a - 2, a - 4$ has non-negative entries for $a_i \geq 4$, so $H^1 = 0$ by Künneth (the same non-negativity as in Proof 2 of Theorem 4.1, with one fewer Koszul step). Thus $\text{def} = 0$ for all $a_i \geq 4$.

3 The rank formula at $b = 2$: a complete proof

Lemma 3.1 (Slices of a generic section). **[proven]** Let $V = H^0(Y, \mathcal{O}(2, 2, 2, 2))$ and $W = H^0(\mathbb{P}^1, \mathcal{O}(2))$.

1. The slice map $H^0(A, \mathcal{O}(2, 2, 2, 2)) \cong V \otimes W \rightarrow V^{\oplus 3}$, $s \mapsto (s_1, s_2, s_3)$, is a linear isomorphism.
2. For (s_1, s_2, s_3) in a dense Zariski-open subset of $V^{\oplus 3}$, the common zero scheme $\mathcal{Z} = \mathcal{Z}(s_1, s_2, s_3) \subset Y$ is finite, and $\dim \langle s_1, s_2, s_3 \rangle = 3$.
3. For such triples, (s_1, s_2, s_3) is a regular sequence on Y ; the Koszul complex

$$0 \rightarrow \mathcal{O}(a - 6) \rightarrow \mathcal{O}(a - 4)^{\oplus 3} \rightarrow \mathcal{O}(a - 2)^{\oplus 3} \xrightarrow{\mu} \mathcal{O}(a) \rightarrow \mathcal{O}_{\mathcal{Z}}(a) \rightarrow 0 \quad (6)$$

is exact, and $\deg \mathcal{Z} = 48$.

Consequently, the slices of a generic $s \in H^0(A, \mathcal{O}(2, 2, 2, 2))$ satisfy (2)–(3).

Proof. (1) is Künneth plus the choice of a monomial basis of W : the coordinates of $s \in V \otimes W$ in that basis are the slices, and $\dim(V \otimes W) = 27 \cdot 3 = \dim V^{\oplus 3}$. (2) $\mathcal{O}(2, 2, 2, 2)$ is very ample on the smooth projective threefold Y ; embed $Y \subset \mathbb{P}^{26}$ by $|V|$. By Bertini (char. 0) [8, II.8.18, III.7.9.1], a generic member of $|V|$ is a smooth surface, a generic second section cuts it in a smooth curve, and a generic third meets that curve in a finite scheme; linear

independence of a generic triple is clear. The locus of triples with $\dim Z \geq 1$ is Zariski-closed (upper semicontinuity of fiber dimension on the incidence variety), so its complement is the required dense open set. (3) Y is smooth, hence Cohen–Macaulay. At points of $\mathcal{Z}$ the three local equations cut a codimension-3 subscheme of a 3-dimensional CM local ring, hence form a regular sequence; away from $\mathcal{Z}$ some s_i is a unit. Exactness of the Koszul complex on a regular sequence is standard [7, Cor. 17.5]. Properness of the intersection gives Bézout on Y : $\deg \mathcal{Z} = \int_Y c_1(\mathcal{O}(2, 2, 2))^3 = 3! \cdot 2^3 = 48$. For generic s the three divisors meet transversally, so $\mathcal{Z}$ is reduced: 48 distinct points, each a complete-intersection point with regular local ring (this is the reducedness used in Theorem 5.1 for the local length count). By (1), generic s yields a generic triple. $\square$

Theorem 3.2 (Rank formula, $b = 2$). *[proven] Let $D = (-a_1, -a_2, -a_3, 2)$, $a_i \geq 4$, s generic. With all cohomology on Y and the convention (2) extended by Künneth,*

$$\text{rank } \alpha = 3 \prod_{i=1}^3 (a_i - 1) - \left[3h^0(a - 4) - h^0(a - 6) + h^1(a - 6) \right], \quad (7)$$

$$\text{def}(D) = \min\left(\prod_i (a_i + 1), 3 \prod_i (a_i - 1)\right) - \text{rank } \alpha. \quad (8)$$

Proof. By Proposition 2.1, $\text{rank } \alpha = \text{rank } \mu$ with μ as in (5), a map from a space of dimension $3h^0(a - 2) = 3 \prod (a_i - 1)$. By left-exactness of H^0 , $\ker \mu = H^0(K_2)$ for the kernel sheaf $K_2 = \ker(\mathcal{O}(a - 2)^{\oplus 3} \rightarrow \mathcal{O}(a))$, which by Lemma 3.1(3) sits in

$$0 \rightarrow \mathcal{O}(a - 6) \rightarrow \mathcal{O}(a - 4)^{\oplus 3} \rightarrow K_2 \rightarrow 0.$$

Its long exact sequence gives

$$h^0(K_2) = 3h^0(a - 4) - h^0(a - 6) + \dim \ker\left(H^1(\mathcal{O}(a - 6)) \rightarrow H^1(\mathcal{O}(a - 4)^{\oplus 3})\right).$$

The decisive vanishing: for $a_i \geq 4$ every entry of $a - 4$ is non-negative, so by Künneth and (2), $H^1(Y, \mathcal{O}(a - 4)) = 0$ *identically on the domain*. Hence the kernel above is all of $H^1(\mathcal{O}(a - 6))$, of dimension $h^1(a - 6)$, and $h^0(K_2)$ equals the bracket in (7). Then $\text{rank } \mu = 3 \prod (a_i - 1) - h^0(K_2)$, which is (7); and (8) is the definition of def with the dimensions of Proposition 2.1 at $b = 2$. $\square$

Remark 3.3 (The minimum flips). *[proven] $\text{src} = \prod (a_i + 1)$ and $\text{tgt} = 3 \prod (a_i - 1)$ satisfy $\text{tgt} < \text{src}$ on the low boundary (e.g. $(4, 4, 4)$: $81 < 125$) but $\text{src} < \text{tgt}$ for all $a_i \geq 6$, since $\prod_i \frac{a_i + 1}{a_i - 1} \leq (7/5)^3 = \frac{343}{375} \cdot 3 < 3$. The defect is insensitive to the flip (Theorem 4.1); the formula (8) handles both regimes uniformly.*

Remark 3.4 (The walls). *[proven] The defect (8) has two distinct sources of piecewise behaviour. First, the h^1 correction terms in the rank formula (7) switch on precisely where an entry of $a - 6$ crosses -2 , i.e. where a single ambient H^1 activates; on the domain $a_i \geq 4$ this happens exactly at entries $a_i = 4$. At $a = (4, 6, 6)$, $h^1(\mathcal{O}(-2, 0, 0)) = 1$ contributes $\text{def} = 28 = 27 + 1$, falsifying any single global cubic. Second, $\text{def} = \min(\dim_{\text{src}}, \dim_{\text{tgt}}) - \text{rank}$ carries the factor $\min(\dim_{\text{src}}, \dim_{\text{tgt}})$, and which argument is smaller flips with the charges (for equal a_i the minimum is $\dim_{\text{tgt}}$ at $a_i \leq 5$ and $\dim_{\text{src}}$ at $a_i \geq 6$); this source-target flip is a separate saturation effect, not an H^1 wall.*

4 The ceiling: two proofs

Theorem 4.1 (Saturation ceiling). *[proven] For all $a_i \geq 6$, $\text{def}(D) = 48$.*

Proof 1 (finite-difference identity). For $a_i \geq 6$ every entry of $a - 6$ is non-negative, so $h^1(a - 6) = 0$, and by Remark 3.3 the minimum is $\text{src} = \prod(a_i + 1)$. Define the cubic polynomial $f(t) = \prod_{i=1}^3(a_i + 1 - 2t)$, so that

$$f(0) = \prod(a_i + 1), \quad f(1) = \prod(a_i - 1), \quad f(2) = h^0(a - 4), \quad f(3) = h^0(a - 6).$$

Then by Theorem 3.2,

$$\text{def} = f(0) - 3f(1) + 3f(2) - f(3) = -\Delta^3 f(0) = -3! \cdot (\text{lead. coeff. of } f) = -6 \cdot (-2)^3 = 48,$$

since the third finite difference of a cubic is constant. This holds identically in (a_1, a_2, a_3) . $\square$

Proof 2 (geometric: the base locus). For $a_i \geq 6$ all twists $a - 2, a - 4, a - 6$ in (6) have non-negative entries, so all their higher cohomology on Y vanishes (Künneth). Chasing the two short exact sequences built from (6): $H^1(K_2)$ is squeezed between $H^1(\mathcal{O}(a - 4)^{\oplus 3}) = 0$ and $H^2(\mathcal{O}(a - 6)) = 0$, hence vanishes, so $H^0(\mathcal{O}(a - 2)^{\oplus 3}) \twoheadrightarrow H^0(\mathcal{I}_{\mathcal{Z}}(a))$, i.e. $\text{im } \mu = H^0(\mathcal{I}_{\mathcal{Z}}(a))$; and $H^1(\mathcal{I}_{\mathcal{Z}}(a))$ is squeezed between $H^1(\mathcal{O}(a - 2)^{\oplus 3}) = 0$ and $H^2(K_2) = 0$, hence vanishes, so $H^0(\mathcal{O}(a)) \twoheadrightarrow H^0(\mathcal{O}_{\mathcal{Z}}(a))$. Therefore

$$\text{coker } \mu \cong H^0(\mathcal{O}_{\mathcal{Z}}(a)) \cong k^{48},$$

the length of the base scheme (Lemma 3.1). The defect equals this cokernel dimension in the saturated regime, independent of the charges: a genuine infinite ceiling. $\square$

Remark 4.2 (Why $3! \cdot 2^3$). *[proven]* Proof 1 explains the numerology: the $3!$ is the third finite difference (three Koszul steps on a threefold), the 2^3 is the cube of the step (N shifts charges by 2). Proof 2 identifies the same number as $\deg \mathcal{Z} = \int_Y (2H_1 + 2H_2 + 2H_3)^3$. The agreement of the combinatorial and geometric counts is the content of the saturation.

5 Higher fourth charge: the partial contraction and a decisive test

For $b \geq 3$ the reduction (4) does not collapse to a single multiplication map: the residual factor W_{b-2}^* has dimension $b - 1 \geq 2$. Computation at $b = 3$ (and spot checks at $b = 4$) shows the defect is governed, between the walls, by the cubic interior law

$$\text{def}(D) = (b + 1) \prod_{i=1}^3 (a_i - (2b - 1)) \tag{9}$$

between the walls, with small boundary h^1 corrections of the same type as Remark 3.4 (e.g. $(-6, -8, -8, 3) \mapsto 38 = 36 + 2$, and $(-8, -8, -8, 3) \mapsto 106 = 108 - 2$), and a deep-region constant ceiling: 192 at $b = 3$, first attained at $(9, 9, 9)$ and stable above it (also at $(10, 10, 10)$, $(8, 9, 10)$), while below it the defect is strictly sub-ceiling — the equal-charge ladder reads 4, 32, 106 at $a_i = 6, 7, 8$. The corrected ceiling criterion is proved below in Theorem 5.1.

5.1 The general Toeplitz saturation theorem

The finite-field computations above are explained by a stronger structural statement. The special section-induced map is not merely expected to have the Thom–Porteous length. Its Toeplitz maximal-minor ideal can be computed exactly for every $b \geq 2$, and the global rank defect is then determined by the source-target dimensions once the Buchsbaum–Rim resolution is acyclic.

Theorem 5.1 (General Toeplitz determinantal saturation). *[proven] Let k be an algebraically closed field of characteristic zero. Let*

$$Y = (\mathbb{P}_k^1)^3, \quad L = \mathcal{O}_Y(2, 2, 2),$$

and let $s_0, s_1, s_2 \in H^0(Y, L)$ be general sections. Set

$$Z = V(s_0, s_1, s_2) \subset Y.$$

Then Z is a reduced complete intersection of length 48.

Fix $b \geq 2$, set $n = b - 1$, and consider the section-induced Toeplitz map

$$\varphi_b : \mathcal{O}_Y^{\oplus(n+2)} \longrightarrow L^{\oplus n},$$

whose local matrix is

$$T_n(x, y, z) = \begin{pmatrix} x & y & z & 0 & \cdots & 0 \\ 0 & x & y & z & \cdots & 0 \\ \vdots & & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & x & y & z \end{pmatrix},$$

with $x = s_0$, $y = s_1$, and $z = s_2$. Let

$$\mathcal{C}_b = \text{coker}(\varphi_b).$$

Then

$$\text{Fitt}_0(\mathcal{C}_b) = \mathcal{I}_Z^{b-1}.$$

Equivalently, the degeneracy scheme of φ_b is the closed subscheme defined by $\mathcal{I}_Z^{b-1}$. Moreover,

$$\text{length}(\mathcal{C}_b) = 48 \binom{b+1}{3}.$$

Let $a = (a_1, a_2, a_3)$ and twist by $\mathcal{O}_Y(a - 2)$, where

$$a - 2 = (a_1 - 2, a_2 - 2, a_3 - 2).$$

The Buchsbaum–Rim complex gives an exact locally free resolution

$$0 \rightarrow \mathcal{O}_Y(a-2b-2)^{\oplus(b-1)} \rightarrow \mathcal{O}_Y(a-2b)^{\oplus(b+1)} \rightarrow \mathcal{O}_Y(a-2)^{\oplus(b+1)} \rightarrow \mathcal{O}_Y(a)^{\oplus(b-1)} \rightarrow \mathcal{C}_b(a-2) \rightarrow 0.$$

If $a_i \geq 2b + 1$ for all i , then all higher cohomology of the line-bundle terms in this resolution vanishes. Hence the induced global-section complex is exact.

Define

$$E = (b+1) \prod_{i=1}^3 (a_i - 1), \quad F = (b-1) \prod_{i=1}^3 (a_i + 1), \quad \mathcal{C}_b = 48 \binom{b+1}{3}.$$

Let

$$\mu_{b,a} : H^0(Y, \mathcal{O}_Y(a-2))^{\oplus(b+1)} \longrightarrow H^0(Y, \mathcal{O}_Y(a))^{\oplus(b-1)}$$

be the global section map, and define

$$\text{def}(\mu_{b,a}) = \min(E, F) - \text{rank}(\mu_{b,a}).$$

Then, for $a_i \geq 2b+1$,

$$\text{def}(\mu_{b,a}) = \begin{cases} C_b, & E \geq F, \\ C_b + E - F, & E < F. \end{cases}$$

In particular, the Thom–Porteous ceiling

$$\text{def}(\mu_{b,a}) = 48 \binom{b+1}{3}$$

holds whenever $a_i \geq 2b+1$ for all i and

$$(b+1) \prod_{i=1}^3 (a_i - 1) \geq (b-1) \prod_{i=1}^3 (a_i + 1).$$

Proof. The decomposition

$$H^0((\mathbb{P}^1)^4, \mathcal{O}(2, 2, 2, 2)) \cong H^0(Y, L) \otimes H^0(\mathbb{P}^1, \mathcal{O}(2))$$

identifies a general tetraquadric section with a general triple $(s_0, s_1, s_2) \in H^0(Y, L)^{\oplus 3}$. Since L is very ample and Y is smooth of dimension three, three general members of $|L|$ cut a reduced complete intersection. Its length is

$$\int_Y c_1(L)^3 = \int_Y (2H_1 + 2H_2 + 2H_3)^3 = 48.$$

It remains to identify the determinantal ideal of the Toeplitz matrix. Work first in $S = k[x, y, z]$. Let $I_n(T_n)$ be the ideal generated by the maximal minors of $T_n(x, y, z)$. Every maximal minor is homogeneous of degree n , hence

$$I_n(T_n) \subseteq (x, y, z)^n.$$

We prove the reverse inclusion by initial terms. Number the columns of T_n by $0, 1, \dots, n+1$. For $0 \leq p < q \leq n+1$, let $\Delta_{p,q}$ be the maximal minor obtained by deleting columns p and q . Fix the weight-refined term order $>$ on S that first compares total y -degree (higher y -degree is larger) and breaks ties by lexicographic order with $x > z$; equivalently it refines the weight vector $(0, 1, 0)$ on (x, y, z) by lex in (x, z) , and is a global monomial order. Expand $\Delta_{p,q}$ as the signed sum over transversals of the banded $n \times n$ submatrix (one nonzero entry per row and per retained column). The transversal $\tau_{p,q}$ taking x in rows $0, \dots, p-1$, y in rows $p, \dots, q-2$, and z in rows $q-1, \dots, n-1$ contributes the monomial

$$\pm x^p y^{q-p-1} z^{n+1-q}.$$

A y occupies the superdiagonal position $(i, i+1)$, so among all transversals $\tau_{p,q}$ is the unique one of maximal total y -degree: scanning the retained columns $\{0, \dots, n+1\} \setminus \{p, q\}$ from the left, the smallest retained column is reachable by exactly one row, which fixes that row's entry, and the assignment then propagates rigidly to the right, so the maximal- y transversal

is unique. Hence $x^p y^{q-p-1} z^{n+1-q}$ occurs in the expansion with coefficient ± 1 and cannot cancel, and it is the unique leading term

$$\text{LT}_>(\Delta_{p,q}) = \pm x^p y^{q-p-1} z^{n+1-q}.$$

Every other transversal has strictly smaller y -degree, hence a strictly $>$ -smaller monomial.

As (p, q) varies with $0 \leq p < q \leq n+1$, the monomials

$$x^p y^{q-p-1} z^{n+1-q}$$

run through all monomials of total degree n in x, y, z . Therefore the initial ideal of $I_n(T_n)$ contains every degree- n monomial. Since $I_n(T_n) \subseteq (x, y, z)^n$, it follows that

$$\text{in}(I_n(T_n)) = (x, y, z)^n,$$

and hence

$$I_n(T_n) = (x, y, z)^n.$$

Let $z_0 \in Z$. Since Z is a reduced complete intersection in the smooth threefold Y , the completed local ring at z_0 is $k[[x, y, z]]$, with $x = s_0$, $y = s_1$, and $z = s_2$ a regular parameter system. The preceding determinantal computation gives

$$I_n(T_n(s_0, s_1, s_2)) = (s_0, s_1, s_2)^n$$

in every completed local ring at a point of Z . Since the completion $\hat{\mathcal{O}}_{Y, z_0}$ of the Noetherian local ring $\mathcal{O}_{Y, z_0}$ is faithfully flat over it, and the formation of Fitting ideals commutes with the flat base change $\mathcal{O}_{Y, z_0} \rightarrow \hat{\mathcal{O}}_{Y, z_0}$, equality of Fitt_0 with $(s_0, s_1, s_2)^n$ after completion implies the same equality in $\mathcal{O}_{Y, z_0}$ itself. Away from Z , at least one of s_0, s_1, s_2 is a unit, and the Toeplitz matrix has full row rank, so the Fitting ideal is the unit ideal there. Hence the stalkwise Fitting ideal is

$$\text{Fitt}_0(\mathcal{C}_b) = \mathcal{I}_Z^n = \mathcal{I}_Z^{b-1}.$$

The length is not inferred from the Fitting ideal alone. It is computed from the Buchsbaum–Rim resolution of the local Toeplitz cokernel. With $\deg x = \deg y = \deg z = 1$, this resolution is

$$0 \rightarrow S(-n-2)^{\oplus n} \rightarrow S(-n-1)^{\oplus(n+2)} \rightarrow S(-1)^{\oplus(n+2)} \rightarrow S^{\oplus n} \rightarrow C_n \rightarrow 0.$$

Therefore

$$H_{C_n}(t) = \frac{n - (n+2)t + (n+2)t^{n+1} - nt^{n+2}}{(1-t)^3}.$$

The module C_n has finite length, so this Hilbert series is a polynomial. Taking the third-order Taylor limit at $t = 1$ gives

$$\text{length}(C_n) = \binom{n+2}{3} = \binom{b+1}{3}.$$

This length is computed in the standard-graded model C_n over $S = k[x, y, z]$, where C_n is supported at the irrelevant maximal ideal (x, y, z) . At each reduced point $z_0 \in Z$ the completed local ring $\hat{\mathcal{O}}_{Y, z_0} \cong k[[x, y, z]]$ is the completion of S at (x, y, z) under $x, y, z \mapsto s_0, s_1, s_2$, and the local Toeplitz cokernel is the completion of C_n ; completion preserves the finite length $\binom{b+1}{3}$. Summing over the 48 reduced points of Z ,

$$\text{length}(\mathcal{C}_b) = 48 \binom{b+1}{3}.$$

The ideal $I_n(T_n) = (x, y, z)^n$ has codimension 3, which equals the expected codimension $(n + 2) - n + 1 = 3$ of the locus where a map of ranks $n + 2 \rightarrow n$ drops rank. By the Buchsbaum–Eisenbud acyclicity criterion ([7, Thm. 20.9]; the Buchsbaum–Rim complex itself is [7, §A2.6]), the Buchsbaum–Rim complex of φ_b is therefore exact. Twisting by $\mathcal{O}_Y(a - 2)$ gives the displayed locally free resolution.

If $a_i \geq 2b + 1$, then every line bundle in this resolution has all multidegrees at least -1 . By Kunneth’s formula on $(\mathbb{P}^1)^3$, such line bundles have no higher cohomology. Therefore taking global sections preserves exactness.

The source and target dimensions of $\mu_{b,a}$ are

$$E = (b + 1)h^0(Y, \mathcal{O}_Y(a - 2)) = (b + 1) \prod_i (a_i - 1),$$

and

$$F = (b - 1)h^0(Y, \mathcal{O}_Y(a)) = (b - 1) \prod_i (a_i + 1).$$

Since global sections are exact and twisting a finite-length sheaf by a line bundle does not change its length,

$$\dim \operatorname{coker}(\mu_{b,a}) = h^0(Y, \mathcal{C}_b(a - 2)) = 48 \binom{b + 1}{3} = C_b.$$

Thus $\operatorname{rank}(\mu_{b,a}) = F - C_b$. If $E \geq F$, then

$$\operatorname{def}(\mu_{b,a}) = F - \operatorname{rank}(\mu_{b,a}) = C_b.$$

If $E < F$, then

$$\operatorname{def}(\mu_{b,a}) = E - \operatorname{rank}(\mu_{b,a}) = E - F + C_b.$$

This proves the piecewise formula and the stated ceiling criterion. $\square$

Corollary 5.2 (Equal-charge saturation threshold). *[proven] Both $\rho(b)$ and the cubic-crossing $\tau(b)$ below are equal-charge thresholds, defined for $a_1 = a_2 = a_3 = d$; for asymmetric charges the ceiling is governed by the multicharge inequality $E \geq F$ of Theorem 5.1, not by τ . For equal charges $a_1 = a_2 = a_3 = d$, the corrected saturation threshold in the acyclic range is*

$$\rho(b) = \min \left\{ d \geq 2b + 1 : (b + 1)(d - 1)^3 \geq (b - 1)(d + 1)^3 \right\}.$$

For $b = 2, 3, 4, 5, 6$ this agrees with the cubic-crossing value

$$\tau(b) = (2b - 1) + \left\lceil 2(b(b - 1))^{1/3} \right\rceil.$$

From $b = 7$ onward the two can differ. At $b = 7$, $\tau(7) = 20$, but

$$E = 8 \cdot 19^3 = 54872, \quad F = 6 \cdot 21^3 = 55566,$$

so $E < F$ and

$$\operatorname{def} = 48 \binom{8}{3} + 54872 - 55566 = 1994,$$

not 2688. The first equal-charge ceiling depth is $\rho(7) = 21$.

Proposition 5.3 (The $b = 4$ finite-field distinction). *[computed] This proposition records a finite-field computation, not a characteristic-zero proof; it corroborates Theorem 5.1 at $b = 4$. Theorem 5.1 and the factorial pattern $(b+1)!2^3$ both retrodict the two ceilings computable by hand: $b = 2$ gives 48 and $b = 3$ gives 192 in either law. The two laws first diverge at $b = 4$: 480 (Thom–Porteous) versus 960 (factorial). Exact rank computation over $\mathbb{F}_p$ distinguishes the two predictions in this computed test. At $a = (12, 12, 12)$ (matrices of order 6655×6591) and $a = (13, 13, 13)$ (8640×8232) the measured defect equals*

$$\text{def} = 480 \quad \text{at both depths,}$$

while the interior law (9) predicts 625 and 1080 respectively — both far above either candidate ceiling. The defect therefore reaches 480 in this computation, with the same value at three independent primes ($p \in \{100003, 100019, 100043\}$) at depth $a = (13, 13, 13)$. This matches the Thom–Porteous number $48[(b-1)^2 + \binom{b-1}{3}] = 480$ and disagrees with the factorial value $(b+1)!2^3 = 960$ in this computed finite-field test.

Remark 5.4 (The $b = 4$ computation in context). *[computed] Proposition 5.3 provides finite-field evidence: the $b = 4$ defect plateaus at 480 (three primes), matching the Thom–Porteous degree $48\binom{5}{3}$ and disagreeing with the factorial value 960. This is the first computed distinction between the two candidate ceilings. It is now a reproducibility check on the general theorem rather than a substitute for proof: Theorem 5.1 supplies the all- b identification in the acyclic range, and Corollary 5.2 gives the corrected equal-charge onset.*

Remark 5.5 (The $b = 5$ computation and the onset). *[computed] A second divergent case verifies both the ceiling law and the source-target transition. At $b = 5$ the candidate ceilings are $48\binom{6}{3} = 960$ (Thom–Porteous) and $6!2^3 = 5760$ (factorial). Exact $\mathbb{F}_p$ rank computation gives $\text{def} = 960$ at $a = (15, 15, 15)$ (two primes, fully logged), matching Theorem 5.1 and missing the factorial value by a factor of six; the deeper point $a = (16, 16, 16)$ returns the same 960 (recorded from a prior two-prime run). The equal-charge ladder 48, 158, 352, 642 at $a = (11, 11, 11), (12, 12, 12), (13, 13, 13), (14, 14, 14)$ is exactly the $C_b + E - F$ branch of the theorem. Saturation first occurs at $a_i = 15$, where $E \geq F$ begins in this case; here $\rho(5) = \tau(5) = 15$.*

Figure 2 shows both regimes directly: in the displayed $b = 3$ and $b = 5$ cases the measured defect climbs through the $E < F$ branch and then locks onto the constant ceiling when $E \geq F$. Figure 3 compares the old cubic-crossing reference $\tau(b)$ with the corrected equal-charge threshold $\rho(b)$; the first discrepancy occurs at $b = 7$.

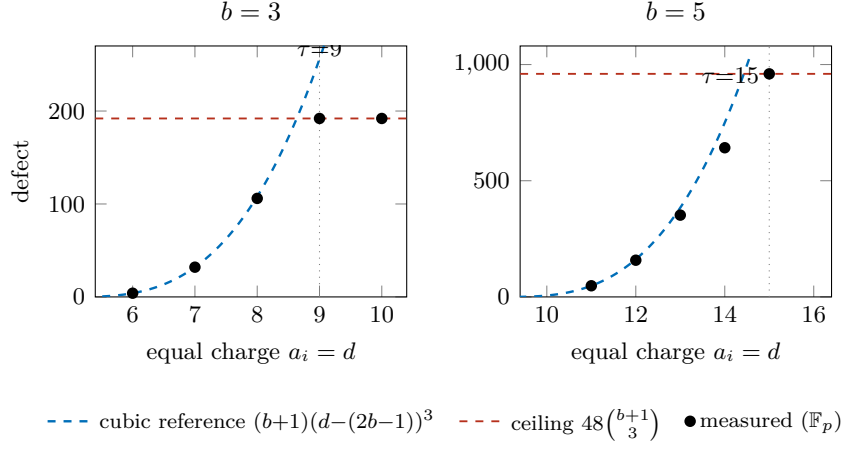


Figure 2: The two regimes of the deep-region defect, with measured values from exact $\mathbb{F}_p$ rank computation (filled markers). In these two low- b cases the corrected equal-charge threshold $\rho(b)$ agrees with the cubic-crossing value $\tau(b)$: $\rho(3) = \tau(3) = 9$ and $\rho(5) = \tau(5) = 15$. Below that point the defect is the $C_b + E - F$ branch of Theorem 5.1; at and above it the defect is the ceiling $48\binom{b+1}{3}$. All plotted values are reproducible from the accompanying repository.

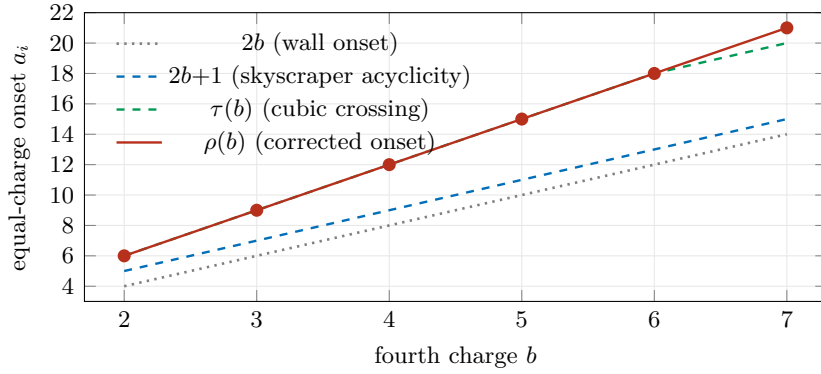


Figure 3: Corrected equal-charge saturation onset. The old cubic-crossing reference $\tau(b) = (2b-1) + \lceil 2(b(b-1))^{1/3} \rceil$ agrees with the corrected threshold $\rho(b)$ through $b=6$, but fails at $b=7$: $\tau(7) = 20$ while $\rho(7) = 21$. The corrected threshold is the first $d \geq 2b+1$ satisfying $(b+1)(d-1)^3 \geq (b-1)(d+1)^3$.

6 Cross-manifold reproduction: the bicubic

On the bicubic $[\mathbb{P}^2 \mathbb{P}^2 \parallel (3, 3)]$ the analogous contraction produces a linear system of plane cubics on $\mathbb{P}^2$. Testing $\mu_V : V \otimes H^0(\mathbb{P}^2, \mathcal{O}(e)) \rightarrow H^0(\mathbb{P}^2, \mathcal{O}(e+3))$, $V \subset H^0(\mathcal{O}(3))$: for a **pencil** ($\dim V = 2$, base locus = complete intersection of two cubics, 9 points by Bézout) the computed defect climbs 1, 3, 6, 9 and saturates at 9 — the base-locus length; for **nets and larger** ($\dim V \geq 3$, empty base locus on a surface) the computed defect is 0 for all $e \geq 4$. This computed bicubic reduction shows the same ceiling-as-intersection-number pattern in a structurally different CICY.

7 Verification

Predictions in the table below are from (7)–(8) with no fitted parameters; measured ranks are exact finite-field computations by generic-coefficient Gaussian elimination over $\mathbb{F}_p$, $p = 2^{31} - 1$, with two random sections per point and fixed seed 20260612. The finite-field computations certify the stated ranks for the sampled finite-field sections and now serve as reproducibility checks against the characteristic-zero theorem, not as substitutes for its proof.

(a_1, a_2, a_3)	pred. rank	meas. rank	pred. defect	meas. defect
(4, 4, 4)	78	78	3	3
(4, 4, 5)	102	102	6	6
(4, 4, 6)	126	126	9	9
(4, 5, 5)	132	132	12	12
(4, 5, 6)	162	162	18	18
(4, 6, 6)	197	197	28	28
(4, 6, 7)	232	232	38	38
(4, 6, 8)	267	267	48	48
(4, 7, 7)	272	272	48	48
(4, 7, 8)	312	312	48	48
(5, 5, 5)	168	168	24	24
(5, 5, 6)	204	204	36	36
(5, 6, 6)	246	246	48	48
(5, 6, 7)	288	288	48	48
(5, 6, 8)	330	330	48	48
(5, 7, 7)	336	336	48	48
(5, 7, 8)	384	384	48	48
(6, 6, 6)	295	295	48	48
(6, 6, 7)	344	344	48	48
(6, 6, 8)	393	393	48	48
(6, 7, 7)	400	400	48	48
(7, 7, 7)	464	464	48	48

22/22 exact. The full X -cohomology satisfies the index identity $\sum_q (-1)^q h^q = 2(\sum_i D_i + \sum_{i < j < k} D_i D_j D_k)$ on every tabulated point, and a 1,728-point sweep over $a \in [4, 15]^3$ finds no violation of (7)–(8) and no defect exceeding 48.

8 Significance and transferability

The content of this computation is not the numerical value on a single Calabi–Yau threefold; it is the isolation of a local determinantal source for a cohomology wall. On the tetraquadric the Koszul section map dualises to a contraction φ_s whose section-induced Toeplitz local model is computed exactly in Theorem 5.1. Its maximal-minor ideal is the $(b - 1)$ st power of the 48-point base-locus ideal, and its local Buchsbaum–Rim cokernel has length $48\binom{b+1}{3}$. In the acyclic range $a_i \geq 2b + 1$, this local length converts into the exact global source-target defect formula. The contribution is therefore not a new cohomology formula, since the dimensions in this region are implicit in the systematised formulae of [2], but a mechanistic derivation: it explains why a wall of this type occurs, why its height carries the binomial factor, and why the old cubic-crossing threshold must be replaced by the source-target inequality.

The most direct use of such a mechanism is as a check on fitted or learned cohomology formulae. Region-by-region polynomial formulae, including those produced by machine-learning methods that identify the Picard-lattice regions and their polynomials and thereby propose a cohomology formula [3, 2], are validated against computed dimensions, yet dimensions alone need not determine the underlying law. The present family makes this concrete: the Thom–Porteous count $48\binom{b+1}{3}$ and the factorial pattern $(b + 1)!2^3$ agree at $b = 2$ and $b = 3$, at 48 and 192, and first diverge at $b = 4$, where 480 separates from 960, and again at $b = 5$, where 960 separates from 5760 (Proposition 5.3, Remark 5.5). A formula calibrated only to the agreeing low-charge values cannot distinguish the two before that first divergence; the determinantal mechanism selects the Thom–Porteous candidate in the computed divergent cases. The cohomology dimensions controlled here are moreover of the kind that enter heterotic spectrum computations on this geometry, the tetraquadric being a standard line-bundle model-building space [1], so the mechanism bears on the interpretability of those computations while making no claim about physical spectra. In the general language of algebraic geometry the wall is a jumping locus realised as the degeneracy locus of a bundle map, its height the length of a determinantal thickening of the base locus (Figure 1); the same local model is expected to govern other complete-intersection families whenever the Koszul connecting map reduces locally to the same Buchsbaum–Rim format, a transfer we treat as a program rather than an established consequence. The bicubic reduction of Section 6 is one instance of that mechanism on a structurally different manifold.

9 Scope, and what is open

The results occupy three levels. The rank formula (Theorem 3.2) and the saturation ceiling at 48 (Theorem 4.1) are proved for $b = 2$ by two independent arguments, identifying the defect with the degree of the base-locus scheme and the third finite difference of the Koszul Hilbert function. The piecewise region formulae whose values this analysis explains are due to Constantin–Lukas [2]; the present contribution is a mechanistic account of one wall family, not a new cohomology formula.

For $b \geq 3$, Theorem 5.1 proves the Toeplitz determinantal mechanism and gives the exact global defect formula in the acyclic range. The exact finite-field rank computations at $b = 3, 4, 5$ now serve as reproducibility checks and as a numerical distinction from the factorial alternative, not as the mathematical proof.

The former universal threshold $a_i \geq \tau(b)$ is false from $b = 7$. The corrected equal-charge threshold is $\rho(b)$ from Corollary 5.2, and the general multicharge ceiling condition is $(b + 1) \prod_i (a_i - 1) \geq (b - 1) \prod_i (a_i + 1)$ together with $a_i \geq 2b + 1$. What remains open is

transferability: whether analogous Toeplitz or Buchsbaum–Rim mechanisms govern higher-codimension complete intersections or sign patterns other than the 3+1 family.

Reproducibility. All rank determinations are exact finite-field linear algebra; no floating point enters. The verification scripts (the rank verifier, the 1,728-point sweep, the Thom–Porteous evaluation, and this table as CSV) are included with this note’s source repository; see the accompanying **README** for runtimes and expected outputs.

Status of claims. **[proven]**: Prop. 2.1, Props. 2.2(1)–(3), Lemma 3.1, Theorem 3.2, Theorem 4.1, Remarks 3.3–4.2, Theorem 5.1, and Corollary 5.2. **[computed]**: the b -law (9), the bicubic reproduction, the table, the $b = 3$ ceiling by exact $\mathbb{F}_p$ rank, the $b = 4$ ceiling at three primes and the $b = 5$ ceiling at two primes (the $b = 2$ ceiling is **[proven]**, Theorem 4.1), the low- b coincidence $\rho(b) = \tau(b)$ for $b = 2, 3, 4, 5$, with $\rho(b)$ the corrected source-target threshold. **[open]**: multi-equation CICYs; non-3+1 sign patterns beyond 2+2; the interior b -law proof for $b \geq 3$ outside the acyclic Toeplitz-saturation range; and transfer criteria for the same Buchsbaum–Rim local model on other complete intersections.

Code and data availability

All rank computations in this note are exact finite-field linear algebra; no floating-point arithmetic enters. The rank formula, the 1,728-point sweep, the Thom–Porteous evaluation, and the data underlying Figures 2–3 are reproducible from the accompanying repository; see the **README** for commands and expected runtimes. The decisive $b = 4$ and $b = 5$ computations require `python-flint` and are the computationally heavy steps.

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